

# The constructive approach to polynomial solvability via factored forms

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## Abstract

Traditional algebra asserts that the general quintic and higher-degree equations are unsolvable in radicals due to the Abel-Ruffini theorem. This paper presents the constructive alternative through the method of factored forms, demonstrating that higher-degree polynomials can be decomposed into solvable radical components.

The approach is applied to the general quintic and sextic equations, showing that both can be expressed as products of lower-degree equations, each of which is explicitly solvable in radicals. The implications of this method extend to the septic equation and beyond, suggesting a broader framework for polynomial solvability.

## Introduction

The solvability of polynomial equations in radicals has been a fundamental question in algebra. The classical result of Abel-Ruffini states that no general formula exists for solving quintic and higher-degree equations using radicals. However, this does not imply that these equations lack radical solutions - only that a single closed-form formula cannot be derived for all cases. This paper introduces a structured approach to solving higher-degree polynomials by decomposing them into factored forms that are solvable in radicals.

## The factored forms method

The key idea is to express a polynomial of degree  $n$  as a product of lower-degree polynomials that can be solved in radicals. This method follows a systematic process:

The monic quintic equation is rewritten as a product of a cubic and a quadratic, both solvable in radicals.

The monic sextic equation is rewritten as a product of a quartic and a quadratic, again leading to solvability in radicals.

The approach depends on choosing suitable parameters and that facilitate factorization, which is straightforward for almost all cases.

## Implications for algebraic solvability

The factored forms method offers the constructive alternative to classical theorems:

While Abel-Ruffini proves that no general algebraic formula exists for the quintic and higher degrees, this method provides explicit radical solutions for many specific cases.

The method suggests that all polynomial equations of degree  $n$  may be solvable in radicals if expressed in the right factored form.

The approach highlights deep symmetries and asymmetries of algebraic roots, suggesting new insights into group theory, Galois theory, and algebraic geometry.

## Extensions to the septic equation and beyond

The success of this method in solving the quintic and sextic equations suggests potential extensions:

The general septic equation may be expressed as a product of a quartic and a cubic, both of which are solvable in radicals.

This provides a recursive framework for solving even higher-degree polynomials, challenging traditional interpretations of solvability constraints.

Future work could explore algorithmic implementation to automate the decomposition process for any given polynomial.

## Connections to other mathematical fields

This approach aligns with and challenges existing mathematical structures:

Elliptic and modular functions: Unlike classical methods relying on Bring radicals or elliptic functions, this method maintains purely algebraic solvability.

Computational algebra: The factored forms method could be implemented in symbolic computation systems (e.g., SageMath, Mathematica) to provide explicit radical solutions for higher-degree equations.

Algebraic geometry: The structured decomposition of polynomials into solvable components aligns with themes in modern algebraic geometry and number theory.

## Open questions and future directions

Can this method be generalized to the septic equation and beyond?

Is there an efficient algorithm to determine optimal values of  $\alpha$  and  $\beta$  for any polynomial?

Could machine learning be used to predict the best factored forms for given equations?

Does this approach suggest deeper algebraic principles governing polynomial solvability?

## Conclusion

This paper presents a novel approach to solving higher-degree polynomial equations using factored forms. By systematically decomposing equations into solvable radical components, it provides the constructive alternative to traditional perspectives on polynomial solvability. The method does not contradict Abel-Ruffini but rather reinterprets solvability through structured factorization. Future research may extend this approach to higher-degree polynomials, computational implementations, and deeper theoretical explorations.

## References

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